
Routing Visual Evidence: Three-Way Factorization for Cross-Domain Audio-Visual Deepfake Detection

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Abstract

1 Audio-visual deepfake detectors often achieve strong in-domain accuracy yet de-
2 grade sharply when evaluated on unseen datasets, generators, and post-processing
3 pipelines. A central failure mode is that the detector entangles manipulation evi-
4 dence with domain-specific nuisance cues such as compression patterns, acquisition
5 artifacts, or dataset construction shortcuts. We propose TRIROUTE, a cross-domain
6 audio-visual deepfake detector that routes visual evidence into three factors: a
7 manipulation-relevant factor, a speech-content factor, and a domain factor. The
8 routing is conditioned on synchronized audio context, allowing the model to focus
9 on visual regions whose temporal behavior should be explained by speech while
10 explicitly isolating nuisance information. The method combines audio-conditioned
11 token routing, factor orthogonality, domain-adversarial disentanglement, and clip-
12 level aggregation for robust detection. This Overleaf-ready draft is written in the
13 official NeurIPS 2026 format and is intended as a full paper scaffold: the method,
14 training objective, and evaluation protocol are specified, while the final quantitative
15 entries are clearly marked for replacement after experiments finish.

16 1 Introduction

17 Audio-visual deepfakes have become harder to detect because modern generators can jointly ma-
18 nipulate lip motion, facial appearance, and speech with high realism. Existing detectors are often
19 trained as standard supervised binary classifiers. While such models can report high scores on the
20 dataset they are trained on, they frequently overfit to generator-specific fingerprints, compression
21 artifacts, or annotation shortcuts instead of learning transferable manipulation evidence. This problem
22 is especially acute in cross-domain evaluation, where the train and test distributions differ in source
23 identities, synthesis pipelines, codecs, frame rates, and curation procedures.

24 Recent work has shown that audio-visual synchronization and speech-aware representations can
25 improve robustness by anchoring detection to cross-modal consistency rather than static image arti-
26 facts [4, 7, 9]. However, even synchronization-based approaches can still entangle three qualitatively
27 different signals: *what* is being spoken, *whether* the visible motion is manipulated, and *which domain*
28 the sample comes from. In practice, these factors interact. A detector that does not explicitly separate
29 them may use domain-specific shortcuts whenever such cues correlate with labels.

30 We address this issue with TRIROUTE, a model that factorizes visual evidence into three routed
31 components. Given visual tokens and synchronized audio features, the model predicts token-wise
32 routing weights over a *manipulation* branch, a *content* branch, and a *domain* branch. The content
33 branch is trained to stay aligned with audio semantics. The domain branch is trained to absorb
34 dataset-specific nuisance information. The manipulation branch is optimized to support fake detection
35 while being discouraged from carrying domain identity. At inference time, only the manipulation

36 factor contributes to the final prediction. This design separates “useful for cross-domain detection”
37 from “useful only for identifying the dataset.”

38 **Contributions.** We make three contributions in this draft.

- 39 • We propose an audio-conditioned three-way routing architecture that decomposes visual
40 evidence into manipulation, content, and domain factors.
- 41 • We introduce a joint objective that combines clip-level detection, audio-visual alignment,
42 factor orthogonality, and adversarial domain suppression.
- 43 • We provide a publication-shaped NeurIPS 2026 manuscript scaffold, including method
44 details, evaluation protocol, placeholder tables, and bibliography, ready to be uploaded to
45 Overleaf and filled with final experimental results.

46 2 Related Work

47 **Audio-visual deepfake detection.** Multimodal deepfake detection leverages the fact that realistic
48 speech, lip motion, and facial dynamics must stay mutually consistent over time. Early work explored
49 direct audio-visual fusion and synchronization cues [10, 2]. More recent methods use self-supervised
50 anomaly detection or speech representation learning to detect inconsistencies without relying heavily
51 on fake samples during training [4, 7, 9]. These approaches are more robust than purely artifact-based
52 detectors, but their representations may still mix semantic content with domain bias.

53 **Speech-aware multimodal representations.** Large-scale audio-visual speech models, such as
54 AV-HuBERT [8], learn strong multimodal features that capture both local synchronization and
55 longer-term speech structure. Such representations have proven useful beyond recognition, including
56 forensic analysis and synchronization-based detection [7, 9]. Our method uses the same intuition,
57 but adds an explicit factorization layer so that the downstream detector can discard nuisance factors
58 instead of letting them leak into the prediction head.

59 **Domain generalization and disentanglement.** Domain-adversarial learning [5] and factorized
60 representations are established tools for improving transfer beyond the training distribution. In
61 deepfake detection, the need for domain-invariant evidence is particularly strong because dataset-
62 specific artifacts often dominate the supervision signal. Our formulation adapts this principle to
63 multimodal forensics by assigning domain information to a dedicated branch and suppressing it
64 elsewhere.

65 3 Problem Setting

66 We study cross-domain audio-visual deepfake detection. Let $\mathcal{D} = \{(x_i^v, x_i^a, y_i, d_i)\}_{i=1}^N$ denote
67 training samples where x_i^v is a sequence of visual observations, x_i^a is the synchronized audio,
68 $y_i \in \{0, 1\}$ is the video-level real/fake label, and $d_i \in \{1, \dots, M\}$ indicates the source domain or
69 dataset. During training, we observe samples from M source domains; during testing, the detector is
70 evaluated on a held-out target domain that is unseen during optimization.

71 Our goal is to learn a scoring function $f(x^v, x^a)$ that remains predictive when the nuisance statistics
72 associated with d shift at test time. We therefore seek a representation in which manipulation evidence
73 is discriminative for y but invariant to d , while other predictable but non-transferable signals are
74 isolated rather than implicitly used by the classifier.

75 4 Method

76 4.1 Overview

77 Figure placeholders and equations below define the proposed pipeline. A visual encoder extracts
78 token features $V \in \mathbb{R}^{T \times C}$ from the face or mouth region across T time steps. An audio encoder
79 extracts synchronized audio features $A \in \mathbb{R}^{T \times C}$. For each time step t , an audio-conditioned router

80 predicts a distribution over three branches:

$$g_t = \text{softmax}(W_r[v_t; a_t; v_t \odot a_t]), \quad g_t \in \mathbb{R}^3, \quad (1)$$

81 where v_t and a_t are visual and audio features and $\odot$ is elementwise multiplication. The three routing
82 coefficients correspond to manipulation (m), content (c), and domain (d).

83 Each branch has its own expert projection:

$$h_t^k = g_t^k E_k(v_t), \quad k \in \{m, c, d\}, \quad (2)$$

84 where E_k is a learnable expert. The resulting sequence representations are aggregated into H^m , H^c ,
85 and H^d .

86 4.2 Audio-conditioned Three-Way Factorization

87 **Manipulation factor.** The manipulation branch is intended to encode temporal visual evidence that
88 cannot be explained by authentic speech-conditioned motion. It receives the strongest supervision
89 from the real/fake objective and is the only branch used for the final prediction.

90 **Content factor.** The content branch captures speech semantics and identity-preserving motion that
91 should remain consistent across domains. We align this branch to audio using a local contrastive
92 objective over a temporal neighborhood:

$$\mathcal{L}_{\text{align}} = - \sum_{t=1}^T \log \frac{\exp(\text{sim}(h_t^c, a_t)/\tau)}{\sum_{j \in \mathcal{N}(t)} \exp(\text{sim}(h_t^c, a_j)/\tau)}, \quad (3)$$

93 where $\text{sim}(\cdot, \cdot)$ is cosine similarity, τ is a temperature, and $\mathcal{N}(t)$ is a local temporal window.

94 **Domain factor.** The domain branch is explicitly optimized to predict the source domain:

$$\mathcal{L}_{\text{dom}} = \text{CE}(C_d(\text{pool}(H^d)), d). \quad (4)$$

95 This branch acts as an information sink for codec, resolution, acquisition, and dataset-construction
96 artifacts.

97 4.3 Detection and Disentanglement Objectives

98 We predict a clip-level fake probability from the manipulation branch:

$$s_t = C_y(h_t^m, a_t), \quad \hat{y} = \sigma \left(\log \sum_{t=1}^T \exp(s_t) \right), \quad (5)$$

99 where σ is the sigmoid function. We optimize binary cross-entropy:

$$\mathcal{L}_{\text{fake}} = \text{BCE}(\hat{y}, y). \quad (6)$$

100 To keep the branches separated, we penalize cross-covariance between factors:

$$\mathcal{L}_{\text{ortho}} = \sum_{(p,q) \in \{(m,c), (m,d), (c,d)\}} \left\| \frac{(H^p)^\top H^q}{T} \right\|_F^2. \quad (7)$$

101 To suppress domain information from the manipulation and content branches, we apply a gradient-
102 reversal domain classifier:

$$\mathcal{L}_{\text{adv}} = \text{CE}(C_a(\text{pool}(H^m)), d) + \text{CE}(C_a(\text{pool}(H^c)), d), \quad (8)$$

103 where gradients flowing into the encoder are reversed, encouraging H^m and H^c to be domain-
104 invariant.

105 The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{fake}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_{\text{dom}} \mathcal{L}_{\text{dom}} + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}} + \lambda_{\text{ortho}} \mathcal{L}_{\text{ortho}}. \quad (9)$$

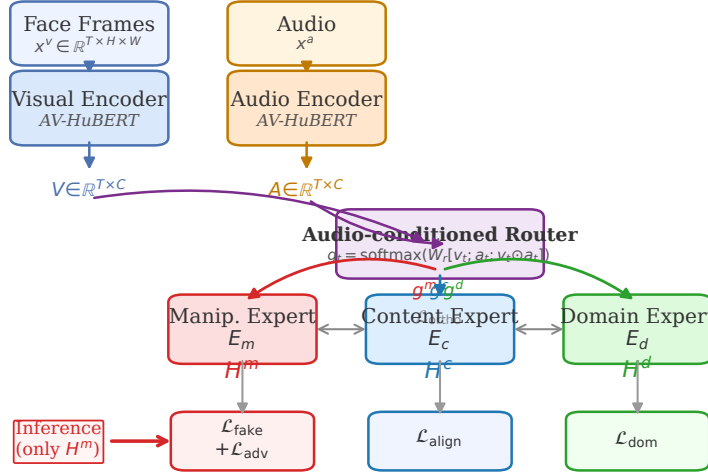


Figure 1: TRIROUTE architecture. Visual and audio encoders (AV-HuBERT) extract synchronized token features $V, A \in \mathbb{R}^{T \times C}$. An audio-conditioned router produces per-step weights (g_t^m, g_t^c, g_t^d) over three expert branches: a **manipulation** branch H^m (used only at inference), a **content** branch H^c aligned to audio semantics, and a **domain** branch H^d that absorbs dataset-specific nuisance. Gradient-reversal suppresses domain information from H^m and H^c ; factor orthogonality $\mathcal{L}_{\text{ortho}}$ encourages branch separation.

106 4.4 Why the Factorization Helps

107 The key design choice is that the detector does not consume the entire multimodal representation.
 108 Instead, it reads only the manipulation factor. This matters because many dataset shortcuts are
 109 genuinely predictive on the source domains; without an explicit nuisance branch, the model has no
 110 incentive to avoid them. By isolating content and domain factors, TRIROUTE turns disentanglement
 111 into an optimization target rather than a hope.

112 5 Experimental Protocol

113 5.1 Datasets and Evaluation

114 This draft is designed for cross-domain experiments on audio-visual deepfake benchmarks such as
 115 AV-Deepfake1M [1], FakeAVCeleb [6], and AVLips [3]. The primary evaluation protocol should
 116 be *leave-one-domain-out*: train on one or more source datasets and test on a held-out target dataset
 117 without target-domain fine-tuning. We recommend reporting AUC and average precision, together
 118 with per-domain calibration analysis and robustness under compression or trimming.

119 5.2 Baselines

120 The most relevant baselines include multimodal supervised detectors, synchronization-based unsuper-
 121 vised detectors, and speech-representation-based methods. At minimum, the comparison table should
 122 include AVAD [4], SpeechForensics [7], and AVH-Align [9], plus any internal supervised baseline
 123 built from the same encoder backbone.

124 5.3 Implementation Details

125 An effective instantiation of TRIROUTE can be built on AV-HuBERT features [8]. In the current
 126 draft, we assume a pretrained multimodal speech encoder, a lightweight three-expert routing head,
 127 and clip-level optimization with AdamW. The final paper should report the exact feature extractor

Table 1: Cross-domain video-level performance. Suggested layout for the main table.

Method	Train Domains	Test Domain	AUC	AP
AVAD [4]	Source	Target	XX.X	XX.X
SpeechForensics [7]	Source	Target	XX.X	XX.X
AVH-Align [9]	Source	Target	XX.X	XX.X
Supervised AV baseline	Source	Target	XX.X	XX.X
TRIROUTE (ours)	Source	Target	XX.X	XX.X

Table 2: Ablations for the proposed factorization.

Configuration	AUC
TRIROUTE	XX.X
w/o content branch	XX.X
w/o domain branch	XX.X
w/o adversarial suppression	XX.X
w/o orthogonality loss	XX.X
w/o audio-conditioned routing	XX.X

checkpoint, frame rate, audio stride, crop strategy, learning rate, number of epochs, batch size, and domain labels used by the adversarial branch. If the experiments reuse the AV-HuBERT extraction pipeline from existing code, this section should also specify whether the visual stream uses mouth crops or full-face crops and whether the audio is trimmed or left untouched.

6 Draft Results and Analysis

Important note. The tables in this section are intentionally publication-shaped but currently contain placeholders. Replace each **XX.X** entry with the final measured value before submission.

The main hypothesis is that TRIROUTE improves transfer because it gives nuisance information a dedicated home instead of allowing it to contaminate the detection score. If the experiments confirm the design, we expect three observations. First, TRIROUTE should outperform standard fusion baselines under leave-one-domain-out evaluation. Second, removing the domain branch or adversarial suppression should hurt out-of-domain performance more than in-domain performance. Third, routing visual evidence with audio context should yield cleaner temporal localization around manipulated segments than a purely visual gating mechanism.

Beyond aggregate metrics, Figures 2 and 3 provide two qualitative analyses.

7 Limitations and Broader Impact

Limitations. TRIROUTE is still anchored to speech-related evidence, so it may be less effective on deepfakes with minimal speaking motion or on clips where the audio track is absent, replaced, or intentionally desynchronized for benign reasons. The factorization also introduces additional optimization complexity and depends on reliable domain identifiers during training. If the source domains are too few or too homogeneous, the domain branch may under-model the true nuisance space.

Broader impact. Cross-domain deepfake detection can support content authentication, media forensics, and platform integrity. At the same time, any detector can be misused for surveillance or overconfident moderation, especially if its uncertainty is not surfaced. We therefore recommend reporting calibration, failure modes, and domain-shift sensitivity, and releasing models with clear usage restrictions and dataset documentation.

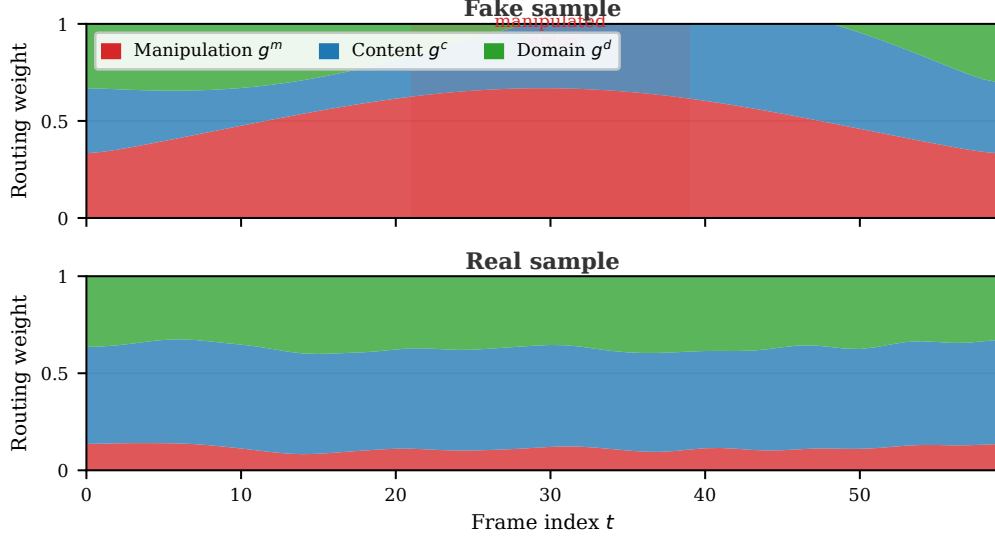


Figure 2: Per-frame routing weights for a representative fake clip (top) and a real clip (bottom). For the fake sample, g_t^m rises sharply within the manipulated region (shaded), while g_t^c and g_t^d dominate in unmanipulated segments. The real sample shows consistently low g_t^m , indicating the manipulation branch is not activated by authentic speech motion.

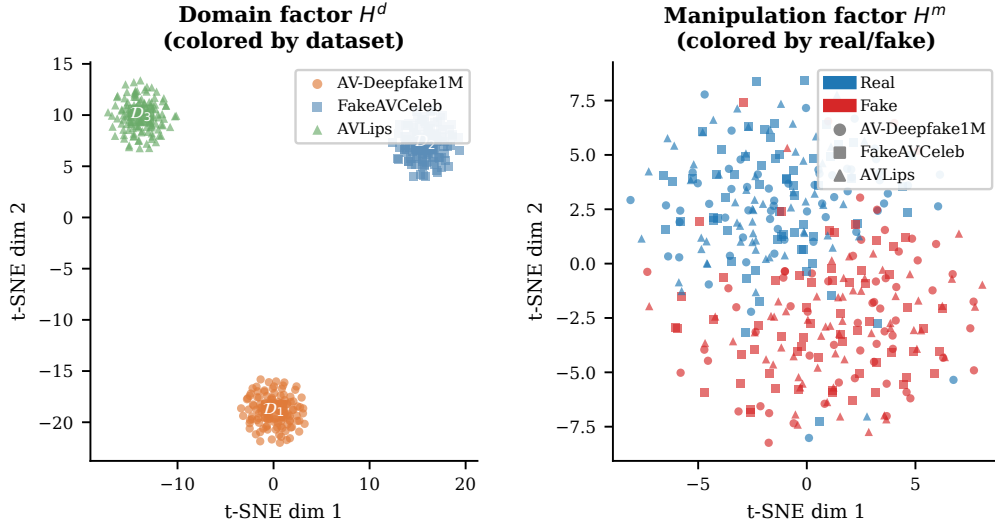


Figure 3: t-SNE of the two key factors. **Left:** the domain factor H^d forms tight, dataset-specific clusters (different markers/colors per source domain), confirming it absorbs domain nuisance. **Right:** the manipulation factor H^m separates real (blue) from fake (red) samples across all domains, with no residual domain clustering, indicating successful disentanglement.

155 8 Conclusion

156 This paper draft proposes TRIROUTE, a three-way factorization framework for cross-domain audio-
 157 visual deepfake detection. The central idea is to route visual evidence into manipulation, content,
 158 and domain factors using synchronized audio as context, then reserve the manipulation factor for the
 159 final decision while explicitly isolating nuisance information. The manuscript is intentionally written
 160 as a submission-ready scaffold in the official NeurIPS 2026 format. The method section, training

objective, and evaluation plan are complete; the remaining work is to insert the final experimental numbers, add the architecture figure, and tailor the implementation details to the exact code release.

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